Applied Data Analysis (CS401)

Lecture 14
ADA in action
17 Dec 2025

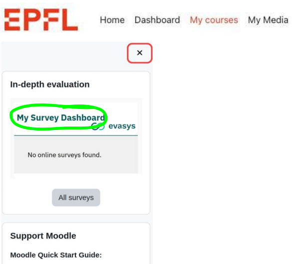
Maria Brbic

Announcements

- Today: last lecture [click here for a famous last lecture]
- No lab session this Friday
- Final project milestone P3 due on **Sun 21 Dec 2025**
- Final exam: **Tue 13 Jan 2026**, 15:15–18:15
  - Dry run, QA example released this week
  - Announcement re: exam protocol, room assignment, etc., to be made in early January

Course evaluation

Course evaluation available on Moodle!



Today we will

show how everything you've learned in lectures 1–13 comes together in one project

Paper available at https://doi.org/10.1073/pnas.2106152118


Postmortem memory of public figures in news and social media


Philip Seymour Hoffman † 2014


Amy Winehouse † 2011

Questions

- Who is remembered by society after death?
  - "Postmortem collective memory"
- Are there prototypical patterns of postmortem collective memory?
- Are certain kinds of people remembered in certain ways?
- Are dead people remembered differently in news vs. social media?

ada #6 Observational studies

Why should we care about postmortem collective memory?

Fact: Humans care a lot about being remembered after death

Ara Pacis, Rome
[Wikipedia]

Damnatio memoriae
[Wikipedia]

An ADA approach

Stuffed Count von Count counting stuff

Let's use lots of data and count stuff!

- **Detect names** of dead people in big corpus of news and social media
- Build time series of name **counts**
- Analyze the **shape of time series**
- **Correlate** shapes with biographic info about dead people from knowledge base

Part 1:
Getting the data

The raw data: spinn3r

- "Spinn3r provides APIs for social media, weblogs, news, video, and live web content to our customers in any language and in large volumes." (Source: spinn3r.com)
- Firehose stored to dis
- Several billions of doc
- 40 terabytes

Postmortem memory of public figures in news and social media

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Edited by Henry L. Roediger III, Washington University in St. Louis, St. Louis, MO, and approved June 24, 2021 (received for review April 1, 2021)

ada #12 Scaling to massive data

Detecting news and social media in Spinn3r

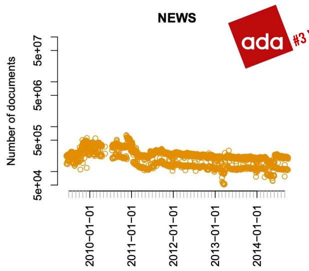
- Found a list of all 151K online news articles about Osama bin Laden's killing (2 May 2011) indexed by Google News
- Assume that every relevant news outlet had reported on bin Laden's death, and that Google News is reasonably complete
- News defined as documents from the 6,608 Web domains appearing in the "bin Laden list"

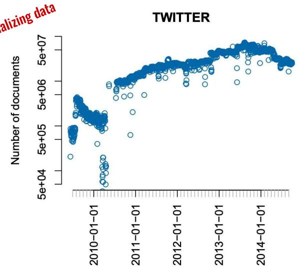
twitter

- Detect via URL pattern: e.g., https://twitter.com/UserName/status/1605097142552403968

ada #2 Handling data

Data volume: Number of docs per day





ada #3 Visualizing data

Detecting mentions of people names

ada #2 Handling data #13 Handling network data #10+11 Handling text data

- Easy: unambiguous names
  - e.g., “Amy Winehouse”
- Hard: ambiguous names
  - e.g., “Michael Jackson”

Wikipedia-based solution:

- Given: a name *X* (e.g., “Michael Jackson”)
- Consider all *N* links in English Wikipedia with *X* as anchor text (cf. examples on the right)
- Build distribution over the *N* link targets
- If there is a target *T* to which ≥90% of all links point: consider *X* “sufficiently unambiguous” and assume that all mentions of *X* in Spinn3r refer to *T*
- Else: consider *X* “too ambiguous” and ignore it in the analysis

The city of Gary is known as the birthplace of singer Michael Jackson.

Beer and whisky expert Michael Jackson was born in Leeds.

...

Recruiting the army of the dead:

	All	Included
Age		
N/A	10%	6%
1st quartile	68	64
Mean	76	74
Median	80	77
3rd quartile	88	87
Gender		
N/A	27%	7%
Female	16%	17%
Male	84%	83%
Manner of death		
N/A	76%	60%
Natural	85%	86%
Unnatural	15%	14%
Language		
N/A	45%	27%
Anglophone	60%	82%
Non-anglophone	40%	18%
Notability type		
N/A	1%	0%
Arts	40%	50%
Sports	14%	14%
Leadership	11%	14%
Known for death	26%	16%
General fame	7%	4%
Academia/engineering	2%	2%
Count	33 340	2 362

- Freebase, a once-popular knowledge graph
  - Now defunct, but still available Freebase
- Contains information about >3M people
  - e.g., Philip Seymour Hoffman = /m/02qgqt
  - More info about some, less about others
- Start from 33K people with death date 2009–2014
  - Further filtering ⟶ 2,362 people
- Extract biographic information from Freebase

ada #2 Handling data #13 Handling network data

Tools used

ada #12 Scaling to massive data hadoop

- Counting names in Spinn3r dump: Hadoop (Java)
- Extracting info from Freebase: Python, Perl
- Once data was small enough: R (script, repo)
  - Statistical analyses
  - Plotting
  - Read data as CSV once (minutes), then serialized to binary format and deserialized in later runs (seconds)

ada #2 Handling data

Our protagonist: Mention time series

- t: relative time t, counting days since death
  - t = 0: day of death
- $S_i(t)$: fraction of documents in which person i was mentioned, out of all documents published on day t
  - For mention time series, consider logarithms: $\log_{10} S_i(t)$

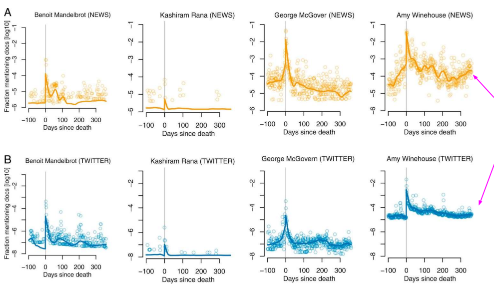
ada #1–13 ADA loves logs!

Mention time series: examples

ada #3 Visualizing data

News

Twitter

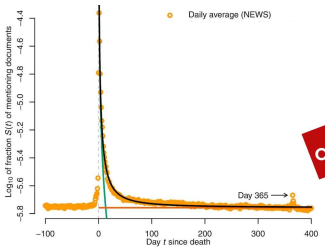


Smoothed via “Friedman’s super smoother”

Part 2:
The shape of postmortem memory

Average mention time series

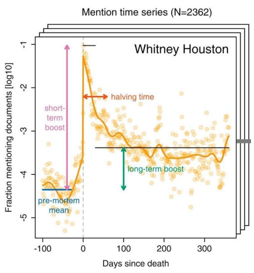
News



#4 Describing data

$\left(\prod_{i=1}^{n} a_i\right)^{\frac{1}{n}} = \sqrt[n]{a_1 a_2 \cdots a_n} = \exp\left(\frac{1}{n}\sum_{i=1}^{n} \ln a_i\right)$

Curve characteristics



Pre-mortem mean: arithmetic mean of days 360 through 30 before death

Short-term boost: maximum of days 0 through 29 after death, minus the premortem mean

Median over people:
1.98
95% CI [1.90, 2.03]

Long-term boost: arithmetic mean of days 30 through 360 after death, minus the premortem mean

Median over people:
0.00055
95% CI [-0.00091, 0.0017]

Halving time: number of days required to accumulate half of the total area between the postmortem curve (including the day of death) and the minimum postmortem value

#4 Describing data

Commercial break


Wikipedia The Free Encyclopedia

Create account Log in

Adenosine deaminase

17 languages

Article Talk Read Edit View history Tools

From Wikipedia, the free encyclopedia

Adenosine deaminase (also known as **adenosine aminohydrolase**, or **ADA**) is an enzyme (EC 3.5.4.4) involved in purine metabolism. It is needed for the breakdown of adenosine from food and for the turnover of nucleic acids in tissues.

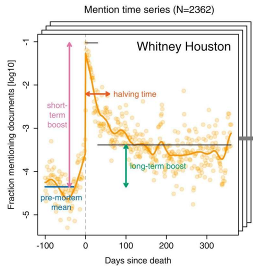
Its primary function in humans is the development and maintenance of the immune system.[5] However, the full physiological role of ADA is not yet completely understood.[6]

ADA

Structure [edit]

ADA exists in both small form (as a monomer) and large form (as a dimer-complex).[6] In the monomer form, the enzyme is a polypeptide chain,[7] folded into eight strands of parallel α/β barrels, which surround a central deep pocket that is the active site.[5] In addition to the eight central β-barrels and eight peripheral α-helices, ADA also contains five additional helices: residues 19-76 fold into three helices, located between β1 and α1 folds; and two antiparallel

Are there prototypical curve shapes?



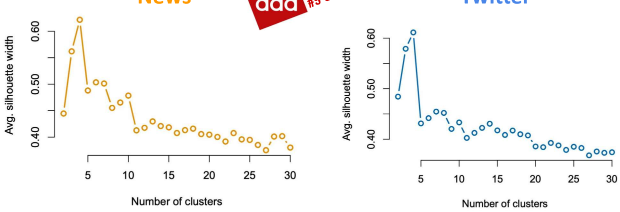
- Each curve one data point
- Represented via its 4 curve characteristics
  - ≈ manual dimensionality reduction
  - Values standardized via z-scores
- Cluster via k-means

#8 Applied ML
#9 Unsupervised learning

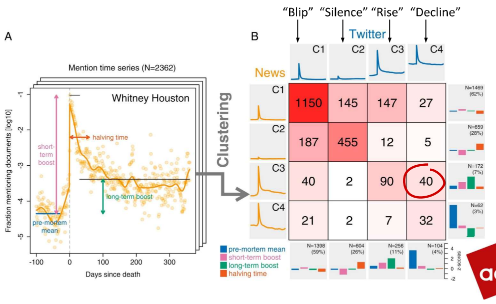
Q: How to find the best number k of clusters?

Average silhouette width!

#9 Unsupervised learning



Cluster analysis



#4 Describing data

Part 3:
Biographic correlates of postmortem memory

A first stab

- Measure correlation coefficient between outcome (e.g., short-term memory boost) and each biographic properties (e.g., gender)
- Higher correlation ⇒ higher outcome for the respective group of people (e.g., women)

Age
N/A
1st quartile
Mean
Median
3rd quartile
Gender
N/A
Female
Male
Manner of death
N/A
Natural
Unnatural
Language
N/A
Anglophone
Non-anglophone
Notability type
N/A
Arts
Sports
Leadership
Known for death
General fame
Academia/engineering
Count

Problem: Biographic properties are correlated

E.g., leaders (politicians, CEOs, etc.) are
- more likely to have died old,
- more likely to have died of a natural death,
- more likely to be men,

compared to artists

ada #5 Regression analysis

Regression analysis allows us to compare averages across subgroups of the data while accounting for correlations among averaged values!

Linear regression

Avg. outcome for "baseline persona": male anglophone artist of median premortem popularity who died a natural death at age 70–79

$$
\begin{aligned}
y_i = \ & \beta_0 \\
& + \beta_1\ \text{premortem\_mention\_freq}_i \\
& + \beta_2\ \text{age\_at\_death}_i \\
& + \beta_3\ \text{manner\_of\_death}_i \\
& + \beta_4\ \text{notability\_type}_i \\
& + \beta_5\ \text{language}_i \\
& + \beta_6\ \text{gender}_i
\end{aligned}
$$

Outcome for person *i*:
- short-term boost or
- long-term boost

Rank-transformed, then linearly scaled/shifted to [–0.5, 0.5]; i.e., median has value 0

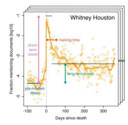
8 discrete levels (dummy-coded): 20–29, 30–39, …, <u>70–79</u>, 80–89, 90–99

2 levels: <u>natural</u>, unnatural

6 levels: <u>arts</u>, sports, leadership, known for death, general fame, academia/engineering

3 levels: <u>anglophone</u>, non-anglophone, unknown

2 levels: <u>male</u>, female

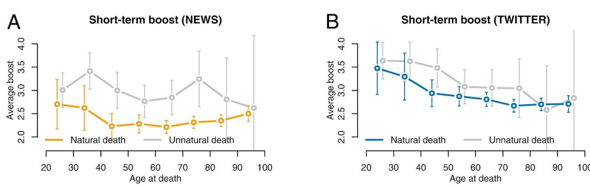


Linear regression results

	Short-term boost (news)	Short-term boost (Twitter)	Long-term boost (news)	Long-term boost (Twitter)
(Intercept)	2.322 (0.063)***	2.670 (0.067)***	0.088 (0.014)***	0.095 (0.015)***
Premortem mean (relative rank)	0.804 (0.093)***	0.948 (0.100)***	0.031 (0.020)	0.086 (0.022)***
Manner of death: unnatural	0.618 (0.095)***	0.282 (0.100)**	0.097 (0.021)***	0.106 (0.022)***
Language: non-anglophone	−0.316 (0.074)***	−0.116 (0.078)	−0.061 (0.016)***	−0.037 (0.017)*
Language: unknown	−0.446 (0.086)***	−0.325 (0.091)***	−0.079 (0.019)***	−0.081 (0.020)***
Gender: female	0.083 (0.072)	−0.034 (0.076)	0.034 (0.016)*	0.006 (0.017)
Notability type: academia/engineering	0.181 (0.197)	0.340 (0.208)	−0.032 (0.043)	0.023 (0.046)
Notability type: general fame	0.070 (0.124)	0.132 (0.131)	−0.010 (0.027)	−0.008 (0.029)
Notability type: known for death	−0.107 (0.099)	−0.088 (0.106)	−0.021 (0.022)	0.008 (0.023)
Notability type: leadership	0.152 (0.083)	0.113 (0.087)	−0.058 (0.018)**	−0.040 (0.019)*
Notability type: sports	0.049 (0.083)	0.072 (0.088)	−0.034 (0.018)	−0.034 (0.020)
Age: 20–29	0.162 (0.170)	0.718 (0.180)***	0.060 (0.037)	0.192 (0.040)***
Age: 30–39	0.400 (0.167)*	0.649 (0.177)***	0.028 (0.037)	0.118 (0.039)**
Age: 40–49	−0.046 (0.126)	0.351 (0.133)**	−0.001 (0.028)	0.100 (0.030)***
Age: 50–59	−0.075 (0.099)	0.181 (0.104)	−0.058 (0.022)**	−0.024 (0.023)
Age: 60–69	−0.109 (0.082)	0.130 (0.086)	−0.050 (0.018)**	−0.025 (0.019)
Age: 80–89	0.022 (0.078)	0.021 (0.082)	−0.018 (0.017)	−0.013 (0.018)
Age: 90–99	0.174 (0.098)	0.034 (0.103)	−0.011 (0.021)	−0.024 (0.023)
R^2	0.213	0.192	0.123	0.178
Adj. R^2	0.197	0.176	0.106	0.161
No. obs.	870	870	870	870
RMSE	0.772	0.815	0.169	0.181

SEs of coefficients are in parentheses. ***$P < 0.001$, **$P < 0.01$, and *$P < 0.05$.

Age at death vs. postmortem memory

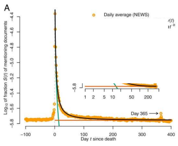


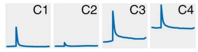
News plays two simultaneous roles (more so than Twitter):
- Catering to public curiosity stirred by a young or unnatural death
- But also when old person or accomplished leader dies

Part 4:
Discussion

Summary: The shape of postmortem memory

- Sharp **pulse** of media attention with death:
  - Median: +9,400% in news, +28,000% on Twitter
- Then sharp **drop** (around 1 month long) toward premortem level
- **Cluster analysis** revealed a set of four prototypical memory patterns: “Blip”, “silence”, “rise”, “decline”
- Same patterns in news and Twitter; **same person** tends to fall into the **same cluster** across the two media





Summary: Biographic correlates

- Notability types: all regression coefficients for long-term boosts negative
  ⇒ All types have lower average long-term boost than default type (artists)
  ⇒ **Artists more present in collective memory**
- **Low R^2** (10–20%): human lives/legacies rich, hard to model
  - But all **model fits highly significant** (*F*-statistic, *p*-value)
  - **Effects** not only significant, but also **large:** e.g., short-term boost (on linear scale) for unnatural vs. natural death: 4x in news, 2x on Twitter
- **Largest boost:**
  - **Premortem popular anglophones who died a young, unnatural death**
  - Long-term boosts **largest for artists, smallest for leaders**